Repeated Measures

PUBH 526

Design and Analysis of Randomized Trials in Public Health

Nilupa S. Gunaratna, MS, PhD

Estimate the Intervention Effect:

Change in intervention group:

70-50=20

Endline only:

70-55=15

Difference-in-difference:

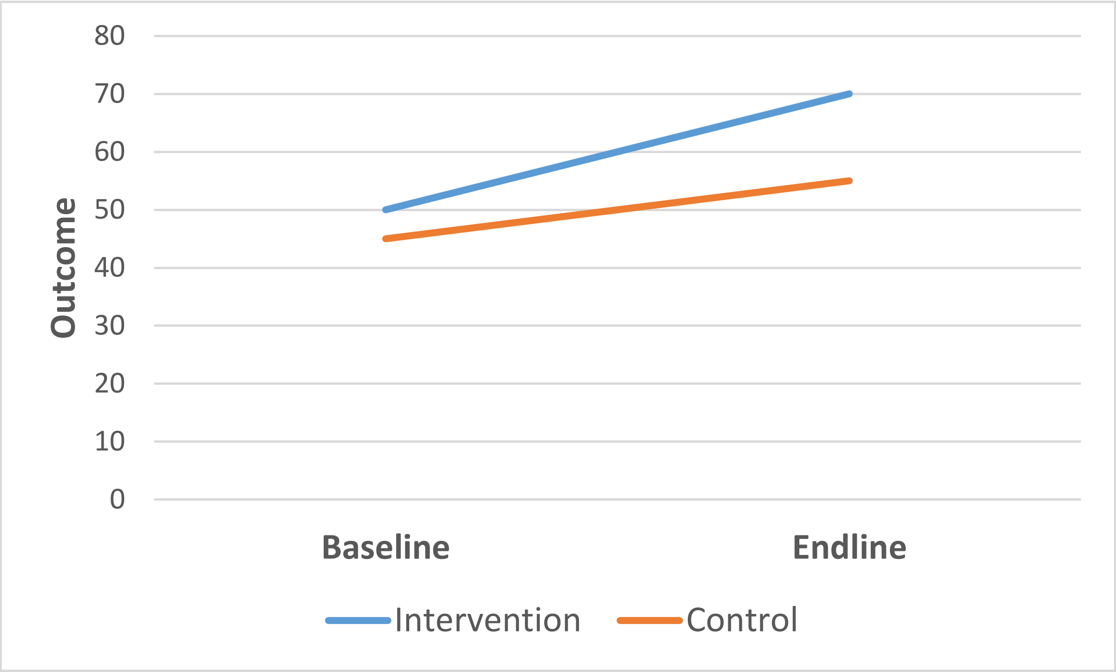
Increase in intervention:

70-50=20

Increase in comparison:

55-45=10

- 10 unit greater increase in the intervention group



	Outcome	
	Baseline	Endline
Intervention	50	70
Control	45	55

Longitudinal Data

- Many RCTs will collect data both before and after intervention (and sometimes at points during the intervention)
  - “baseline”, “endline”, and sometimes “midline”

- Repeated measures on a study participant are not independent – we need to account for this correlation

Options

1. Analyze endpoint data only
    - We know/assume there are no baseline differences on average because we randomized
    - Why did you collect data at baseline then?

2. Use the change in your outcome variable as the response:
    - For every participant, their new $Y = Y_{endline} - Y_{baseline}$
    - Then you only have 1 observation per participant
    - Will there be a constant effect of the treatment on the change, regardless of initial value?

Options

3. Model Y longitudinally, and the treatment x time interaction indicates the treatment effect
   - This is equivalent to a difference-in-difference estimate

4. ANCOVA: $Y_{endline}$ is your response, and $Y_{baseline}$ is a predictor
   - CAUTION: This answers a slightly different question

Difference-in-Difference

- One way to look at treatment effects over the course of a study (i.e., over time)
- Method also used in non-randomized studies